

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Worked Solutions

Applications of Logarithms

Q001 Written

Solve $\log_2 x = 4$.**Solution**

1. Set the argument equal to 2^4 ; hence $x = 16$.

Final answer

16

Q002 Written

Solve $\log_2(2x - 1) = \log_2 3$.**Solution**

1. Domain: $2x - 1 > 0$. Equal arguments give $2x - 1 = 3$, so $x = 2$.

Final answer $x = 2$

Worked Solutions

Applications of Logarithms

Q003 Written

Solve all 12 source logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Solution

1. Write every argument condition, combine logs, solve the resulting algebraic equation, and verify each candidate.

Final answer

Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.

Q004 Written

Solve $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ completely.

Solution

1. Domain gives $x > 2$. Convert $\log_9$ to base 3, solve $(x + 2)^2 / (x - 2)^2 = 9$, and retain only $x = 4$.

Final answer

$x = 4$

Worked Solutions

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.**Solution**

1. State the domain, preserve the sign for base 3, reverse it for base $1/2$, then intersect.

Final answer

$$0 < x < 81; \quad 5 < x < 6$$

Q006 Written

Solve the six source logarithmic inequalities using base monotonicity and argument conditions.**Solution**

1. For $a > 1$ keep the inequality direction; for $0 < a < 1$ reverse it before solving.

Final answer

Each solution is the intersection of the argument domain with the monotonicity interval.

Worked Solutions

Applications of Logarithms

Q007 Written

Solve all 24 source logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Solution

1. Never omit the positive-argument condition; it can remove one branch of an algebraic solution.

Final answer

Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.

Q008 Written

Solve the source challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Solution

1. Combine the logarithms, solve the rational inequality, and retain only the common domain.

Final answer

Reverse the comparison after converting to a base below one, then intersect with both argument conditions.

Worked Solutions

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16\log_2 x + 3 < 0$.**Solution**

1. Let $t = \log_2 x$: $(5t - 1)(t - 3) < 0$, so $1/5 < t < 3$, then back-substitute.

Final answer

$$2^{1/5} < x < 8$$

Q010 Written

Solve the three source substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.**Solution**

1. The substitution turns the problem into a quadratic; only then return to the original logarithm.

Final answerSubstitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.

Worked Solutions

Applications of Logarithms

Q011 Written

Solve the source groups: three equations, three inequalities, and three exact logarithmic values.

Solution

1. Record the domain before setting $t = \log_a x$, and verify every returned value.

Final answer

All groups follow the same substitution, domain, solve, and back-substitution sequence.

Q012 Written

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ for the positive wave-intensity multiplier x .

Solution

1. Let $t = \log_3 x$: $(t - 3)(t + 2) < 0$, so $-2 < t < 3$, hence $3^{-2} < x < 3^3$.

Final answer

$1/9 < x < 27$

Worked Solutions

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Solution

1. Let $t = \log_3 x$, so $0 \leq t \leq 3$; complete the square $y = 7 - (t - 2)^2$.

Final answer

Maximum 7 at $x = 9$; minimum 3 at $x = 1$.

Q014 Written

Find the maximum and minimum of the source logarithmic quadratic on its stated interval, with the corresponding values of x .

Solution

1. Convert the x -interval to a t -interval before reading the quadratic range.

Final answer

Use the vertex in $t = \log_a x$ and compare it with both endpoints.

Worked Solutions

Applications of Logarithms

Q015 Written

Solve the three source extrema problems, each with a different base and interval.

Solution

1. The maximum/minimum in t must be translated back to the exact $x = a^t$.

Final answer

For every item, map the domain to t , locate the vertex, and check endpoints.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Solution

1. Here $0 \leq t \leq 3$ and $y = 5 - (t - 1)^2$; compare $t = 0, 1, 3$.

Final answer

Maximum 5 at $x = 2$; minimum 4 at $x = 1$.

Worked Solutions

Applications of Logarithms

Q017 Written

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Solution

1. Take common logs: $30(\log 2 + \log 3) = 23.343$ and $50(\log 2 - \log 3) = -8.805$.

Final answer

24 digits; 9th decimal place.

Q018 Written

Complete the source digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Solution

1. Use the larger absolute exponent for a number between two negative powers of ten.

Final answer

Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.

Worked Solutions

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Solution

1. Compute $50 \log_{10} 12 = 50(2 \log 2 + \log 3) = 53.965$; then use the characteristic and mantissa.

Final answer

55 digits; leading digit 1.

Q020 Written

Solve $\log_x(x+2) \leq \log_x(4-x)$ for all real x .

Solution

1. Domain: $0 < x < 4, x \neq 1$. For $0 < x < 1$ reverse the sign and for $1 < x < 4$ keep it; both cases are impossible.

Final answer

No real solution.

Worked Solutions

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Solution**

1. A logarithm argument must be positive.

Correct option: C**Final answer**

$$x > 5$$

Q022 MCQ

If $\log_2(2x - 1) = \log_2 3$, then $x =$:**Solution**

1. Equate same-base arguments.

Correct option: B**Final answer**

$$2$$

Worked Solutions

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Solution**

1. The domain $x > 2$ rejects the other algebraic root.

Correct option: C**Final answer**

4

Q024 MCQ

For same-base logarithms, equality of logs gives:**Solution**

1. The logarithm is one-to-one on its domain.

Correct option: A**Final answer**

arguments equal

Worked Solutions

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:**Solution**

1. The logarithm is increasing.

Correct option: B**Final answer**

$$M < N$$

Q026 MCQ

For base $0 < a < 1$, the inequality direction is:**Solution**

1. A reciprocal-base logarithm is decreasing.

Correct option: B**Final answer**

reversed

Worked Solutions

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Solution**

1. The argument must be positive.

Correct option: C

Final answer

$$x > 5$$

Q028 MCQ

The warm-up intersection is:**Solution**1. Intersect the algebraic interval with $x > 5$.

Correct option: B

Final answer

$$5 < x < 6$$

Worked Solutions

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Solution**

1. It turns the logarithm into a single variable.

Correct option: B**Final answer**

$$t = \log_a x$$

Q030 MCQ

The inequality $5t^2 - 16t + 3 < 0$ gives:**Solution**

1. Factor and use the sign chart.

Correct option: B**Final answer**

$$1/5 < t < 3$$

Worked Solutions

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Solution**

1. A logarithm value can be any real number.

Correct option: B**Final answer**

$$t \in \mathbb{R}$$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:**Solution**

1. Convert $-2 < t < 3$ back with base 3.

Correct option: B**Final answer**

$$1/9 < x < 27$$

Worked Solutions

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Solution

1. The vertex gives the maximum.

Correct option: C

Final answer

$$t = 2$$

Q034 MCQ

The correct first step for extrema is:

Solution

1. Then map the domain to a t-range.

Correct option: B

Final answer

$$\text{set } t = \log_a x$$

Worked Solutions

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Solution**1. Apply $\log_3$ to the endpoints.

Correct option: B

Final answer $[0, 3]$

Q036 MCQ

The water-tank challenge maximum health rating is:**Solution**1. The vertex is at $t = 1$, so $x = 2$.

Correct option: C

Final answer

5

Worked Solutions

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Solution**

1. Use the characteristic of the common logarithm.

Correct option: B**Final answer**

$$[\log_{10} N] + 1$$

Q038 MCQ

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:**Solution**

1. Use the larger absolute negative exponent.

Correct option: B**Final answer** 9^{th}

Worked Solutions

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Solution**

1. A common logarithm has base 10.

Correct option: C**Final answer**

10

Q040 MCQ

The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:**Solution**

1. The increasing and decreasing cases both contradict their domains.

Correct option: D**Final answer**

empty

Worked Solutions

Applications of Logarithms

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Solve $\log_5(x - 1) = 2$.

Solution

1. $x - 1 = 25$.

Correct option: A

Final answer

$$x = 26$$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $0 < a < 1$ **and** $M < N$, **compare** $\log_a M$ **and** $\log_a N$.

Solution

1. The function is decreasing.

Correct option: B

Final answer

$$\log_a M > \log_a N$$

Worked Solutions

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.

Solution

1. Let $t = \log_2 x$: $(4t - 1)(t - 1) = 0$.

Correct option: C

Final answer

$$x = 2, \sqrt{2}$$

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2\log_3 x + 4$ **on** $1 \leq x \leq 27$.

Solution

1. Let $t = \log_3 x \in [0, 3]$; vertex at $t = 1$.

Correct option: D

Final answer

$$5 \text{ at } x = 3$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.**Solution**

1. $x + 1 = 9$.

Final answer

$$x = 8$$

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.**Solution**

1. Base 2 is increasing.

Final answer

$$x \geq 8$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $(\log_5 x)^2 - 4 < 0$.**Solution**

1. $-2 < \log_5 x < 2$.

Final answer

$$1/25 < x < 25$$

Quiz WRITTEN 8 Concept Quiz · Written**How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?****Solution**

1. $20(0.3010 + 0.4771) = 15.562$.

Final answer

16 digits

Answer key

Use the question number shown on each page.

- Q001** 16
Q002 $x = 2$
Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
Q004 $x = 4$
Q005 $0 < x < 81$; $5 < x < 6$
Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
Q009 $2^{1/5} < x < 8$
- Q010** Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
Q012 $1/9 < x < 27$
Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
Q017 24 digits; 9th decimal place.
Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019** 55 digits; leading digit 1.
Q020 No real solution.
Q021 C
Q022 B
Q023 C
Q024 A
Q025 B
Q026 B
Q027 C
Q028 B
Q029 B
Q030 B
Q031 B
Q032 B
Q033 C
Q034 B
- Q035** B
Q036 C
Q037 B
Q038 B
Q039 C
Q040 D
Quiz MCQ 1 A
Quiz MCQ 2 B
Quiz MCQ 3 C
Quiz MCQ 4 D
Quiz WRITTEN 5 $x = 8$
Quiz WRITTEN 6 $x \geq 8$
Quiz WRITTEN 7 $1/25 < x < 25$
Quiz WRITTEN 8 16 digits